A Point Cloud of Languages

A reconstruction-free dissimilarity from inferred consonantal correspondences, and the branch structure legible in it — Indo-European, with an Austronesian contrast

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Abstract

We define a dissimilarity between two documented language systems that uses no reconstruction, no family tree, and no branch labels. From concept-aligned wordlists we detect coderivative sets statistically (LexStat), align each language pair, and average — over aligned consonant slots — the number of primary phonological features that differ. Embedding this dissimilarity (classical MDS) and joining nearest neighbours yields a point cloud in which, on 50 Indo-European doculects, a system's nearest neighbour shares its branch 97.7% of the time (silhouette +0.34). A label-permutation test on the fixed matrix puts both figures far above chance ($p \approx 0.0001$; chance purity 0.16), and the result is robust to macro-averaging, to removing 15 near-duplicate systems, and to the distance threshold, and is uncorrelated with coverage. We are deliberately careful about what this shows. Branch recovery is not a novelty of the method: a plain edit distance on consonant skeletons matches our purity, and an unfiltered all-concept baseline matches or beats it — evidence that the branch signal is generic lexical-phonological similarity rather than an artefact of the coderivation filter. The contribution is instead representational: a reconstruction-free, feature-decomposable dissimilarity — the system-level face of the operator repertoire studied in the pilot — whose geometry makes branch structure legible without positing ancestors. Built as a separate field, Austronesian (45 doculects) reproduces the method with a markedly weaker, higher-dimensional signal (purity 0.80, silhouette +0.13); comparing the two clouds' *abstract structure* — never a shared space — suggests that these signatures may fingerprint how a family diversified. Cross-branch proximities are rare and only marginally interpretable as areal contact; we do not claim to identify contact.

1. Introduction

The pilot study argued that a family's inventory of feature-difference operators, read at the level of *types*, is representational rather than genealogical: the operators are largely those general phonology makes available, and the genealogical signal lives one level up, in how those operators are *distributed* across languages. This paper looks at the simplest projection of that distribution — a pairwise dissimilarity between whole systems — and asks a deliberately modest question: is branch structure legible in it, built without any historical apparatus?

The stance is the programme's: discover first, contrast later. We construct the map from correspondences in concept-aligned lexical data and consult branch labels, contact history, and baselines only afterwards. We say coderivative, not *cognate*: operationally, an *algorithmically linked form set* (recurrent correspondence), with no claim of single-ancestor descent. A language, on this view, is a confluence, not the linear issue of one parent; branch structure is one legible aspect of the geometry, not the whole of it.

Two honesty commitments frame the paper. First, the input is not "phonology alone": it is phonological dissimilarity computed *within concept-matched, statistically inferred lexical correspondences.*

"No reconstruction", "no tree", and "no branch labels" are the defensible claims. Second, recovering known branches is a sanity check, not the contribution — §5 shows simpler baselines do as well or better. What is new is the reconstruction-free, feature-decomposable *representation* and its geometry, and (§7) the comparison of that geometry's abstract shape across families held in separate fields.

2. A dissimilarity between systems

Let $\phi_f(x) \in \{+, -, 0\}$ be primary feature f of segment x (panphon). For aligned segments a, b,

$$\mathrm{fd}(a, b) = \left|\{f \in \mathrm{PRIM} : \phi_f(a) \neq \phi_f(b)\}\right|,$$

over the twelve consonantal features `cont, voi, nas, ant, cor, lab, back, round, strid, hi, lo, son`.

Coderivative sets are detected with LexStat (no etymologies, protoforms, or family labels supplied). Within each set, each language pair is aligned by a feature-cost Needleman–Wunsch, and we average fd over slots where both aligned segments are consonants:

$$d(\ell, \ell') = \mathrm{mean}\{\mathrm{fd}(a, b) : (a, b) \text{ a shared consonant slot of } \ell, \ell'\},$$

requiring at least `MINSLOT` slots. Its observed range here is $\approx [0.5, 3.5]$ (theoretical $[0, 12]$).

We call d a dissimilarity, not a metric. Because each pair is computed on its own set of shared slots, the triangle inequality and identity-of-indiscernibles are not guaranteed. It is symmetric and non-negative; that is all we assume. This matters for §3.

3. From dissimilarity to picture

We embed $D = [d(\ell, \ell')]$ with classical (Torgerson) MDS and build a $k = 3$ nearest-neighbour network. Two distinct objects must not be confused: purity is a $k = 1$ statistic (is the single closest system same-branch?); the drawn network uses $k = 3$. The plane is only a projection: on the Indo-European sample the first two MDS axes carry 27.5% of the variance (eigenvalues 38.2, 28.9, 26.0, 17.3, 13.0, ...), and the Gram matrix has 6.7% negative inertia — so D is only *approximately* Euclidean and the 2-D figure is a rough, lossy shadow. All statistics below use the full matrix, not the plane.

4. Results (Indo-European, 50 doculects, 4 creoles)

Branch assignment (for scoring only) follows Glottolog.

Significance. A label-permutation test on the fixed matrix (10,000 permutations of branch labels over the 46 non-creole systems) gives purity 0.977 against a null of 0.155 ± 0.067 (p $\approx$ 0.0001) and silhouette $+0.338$ against -0.190 ± 0.035 (p $\approx$ 0.0001). The geometry aligns with branches far beyond chance.

Purity, honestly.

Denominator	purity
All doculects (multi-member branches)	0.958 (46/48)
Multi-branch, creoles excluded	0.977 (43/44), Wilson 95% [0.88, 1.00]
Macro-average by branch	0.935
After collapsing near-duplicate pairs ($d < 1.3$; 15 dropped)	0.966 (28/29)

That purity stays at 0.966 after removing fifteen near-twin systems (dialect pairs, doculect variants) shows the result is not manufactured by a handful of near-identical pairs. The Wilson interval is wide and the observations are not independent (mutual neighbours; small clusters) — the honest statement is ”far above chance”, not ”a precise 0.98”.

Coverage is not the driver. Per pair, the median count of consonant slots is 676 (range 144–7227) over a median 236 shared concepts; `corr(d, #slots) = +0.01`.

Systems that are near-twins remain each other’s nearest neighbour (e.g. Spanish–Portuguese, Croatian–Serbo- Croatian, Armenian E.–W.), and singleton branches sit next to their plausible relatives rather than far from everything (Armenian nearest Greek/Indo-Iranian) — being alone in the sample is a fact of sampling, not isolation.

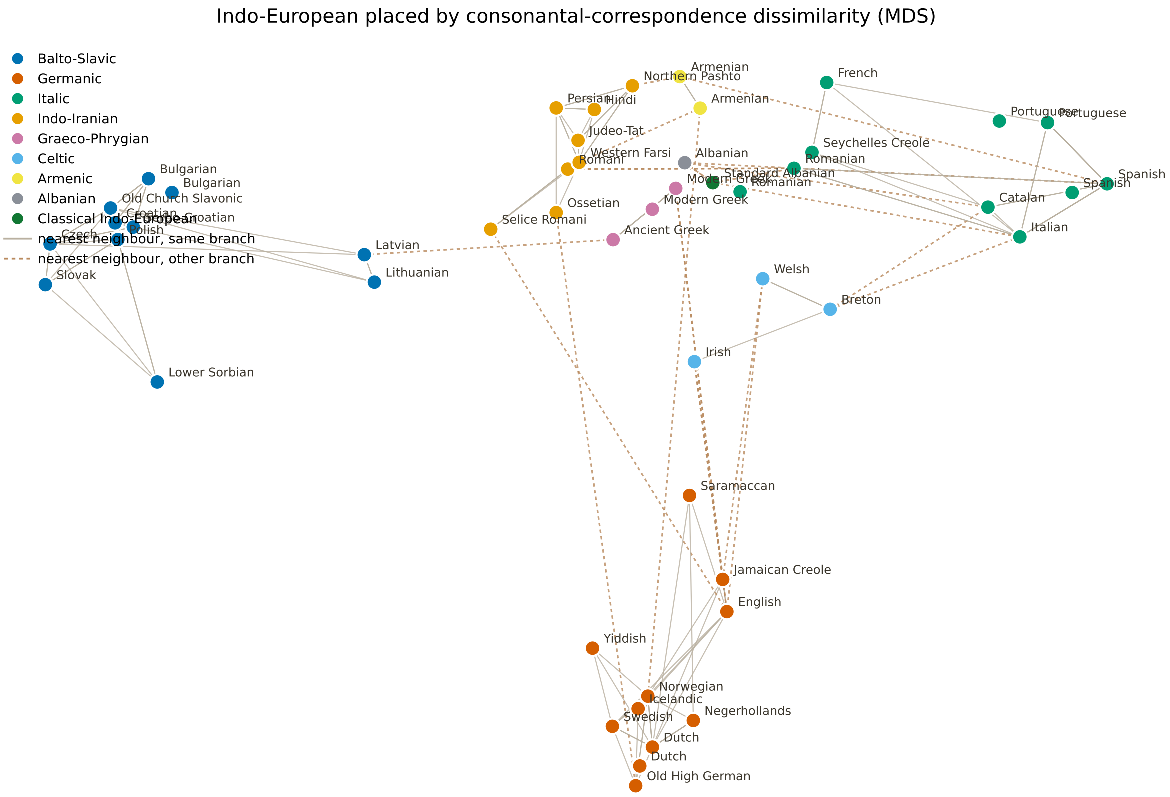

Figure 1: Indo-European doculects placed by the consonantal-correspondence dissimilarity (classical MDS; the plane shows 27.5% of the variance, so central overlaps are partly projection). Nodes coloured by branch — colours added after layout. Solid links join same-branch nearest neighbours; dashed links join the few cross-branch nearest neighbours.

5. Baselines — what the representation does and does not add

On the multi-branch non-creole set (n = 44):

Method	purity	silhouette
Per-language marginal profile	0.477	−0.104
Edit distance on consonant skeletons	0.977	+0.205
All-concept dissimilarity, no LexStat filter	1.000	+0.324
Ours (pairwise feature dissimilarity)	0.977	+0.338

Three honest readings. (i) Branch recovery is generic: a plain edit distance matches our purity, and an unfiltered all-concept dissimilarity matches or beats it. Recovering branches is easy lexical-phonological similarity; we claim no purity novelty. (ii) But our dissimilarity separates the branches more cleanly than edit distance (silhouette +0.34 vs +0.21) — the feature representation buys margin, not just parity. (iii) The result is not a double-dipping artefact: removing the similarity-based coderivation filter (the all-concept baseline) *strengthens* the structure, so the signal is not manufactured by LexStat's selection; the marginal baseline (0.477) confirms the pairwise comparison keeps information a per-language profile discards.

What our representation adds over edit distance is margin plus decomposability and provenance: d is reconstruction-free and factors into *which features change* — the same feature-operators of the pilot — so the point cloud is the system-level face of that operator repertoire, not an opaque string distance.

6. Cross-branch proximities — rare, and barely areal

Of the 50 systems, only four have a nearest neighbour in another branch, and two of those are an artefact:

System	branch	nearest neighbour	branch	d	note
Romani	Indo-Iranian	Romanian	Italic	2.48	Balkan / Romania contact
Irish	Celtic	Jamaican Creole	Germanic	2.76	no obvious story
Albanian (Tosk)	Albanian	Standard Albanian	(Glottolog split)	0.79	same language, label artefact
Standard Albanian	(split)	Albanian (Tosk)	Albanian	0.79	same language, label artefact

Only one (Romani $\rightarrow$ Romanian) matches a documented contact situation; one is a Glottolog labelling split of a single language, and one (Irish $\rightarrow$ Jamaican Creole) has no plausible contact explanation and most likely reflects the projection and sample. We therefore make no areal claim: the geometry does not, on this evidence, independently identify contact or separate it from descent. A proper test requires a contact matrix defined *before* inspecting the map, then asking whether residual proximity tracks it — deferred to future work.

7. A cross-family contrast, in separate fields

We built a second cloud for Austronesian (45 doculects) with the identical pipeline. Crucially, the two families are never placed in a shared distance space — doing so would assert a direct relation we do not claim. We compare only *scale-free structural signatures* of each independently-built matrix: the MDS spectrum (effective dimensionality via the participation ratio; variance in 2-D), branch separation (within/between mean dissimilarity; silhouette; purity), and the self-normalised dissimilarity distribution. No coordinate alignment, no Procrustes.

Field (separate)	n	eff. dim. (PR)	var. in 2-D	within/between	silhouette	purity
Indo-European	50	14.2	27.1%	0.63	+0.32	0.96
Austronesian	45	16.8	22.1%	0.75	+0.13	0.80

Indo-European's branch geometry is sharper and lower-dimensional (tighter within-branch relative to between, higher silhouette, fewer effective dimensions) than Austronesian's, whose dissimilarities cluster near their median — languages nearly equidistant, a flatter and more diffuse cloud (Figure 2). This is a statement about the *shape* of two independent geometries, not about any relation between the families. We record, but do not test here, the hypothesis that such a signature may fingerprint a family's mode of diversification.

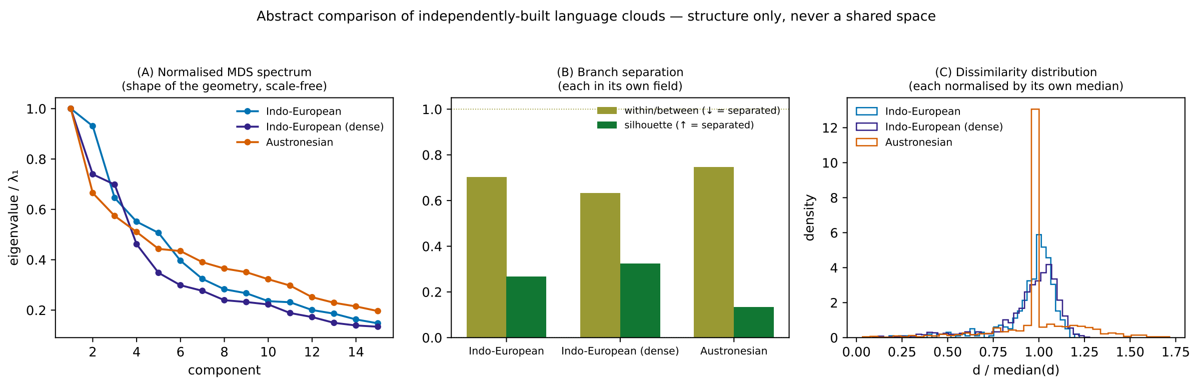

Figure 2: Abstract comparison of the two independently-built clouds — structure only, never a shared space. (A) normalised MDS spectra; (B) branch separation; (C) each cloud's dissimilarity distribution normalised by its own median.

8. Controls and robustness

The label-permutation test (§4) is the primary significance check. In addition: purity and silhouette are unchanged across `MINSLOT` $\in \{20, 30, 40, 60\}$ and across feature subsets (dropping stridency, dropping all place features, or keeping only manner); subsampling systems keeps purity high; and a concept-permuted ablation — shuffling which forms share a concept and rerunning the full pipeline — collapses the structure toward chance, confirming it needs real lexical pairings (reported qualitatively, not as a σ).

9. Limitations and scope

The input is concept-aligned lexical data, not phonology alone; the dissimilarity is corpus- and aligner-dependent and is not a metric; the 2-D figure shows about a quarter of the variance and D is only approximately Euclidean (6.7% negative inertia). Branch recovery is not distinctive in purity versus edit distance (though our silhouette is higher). The areal reading is at most one case. Only consonants are used; the sample is mixed-provenance (see the manifest, Appendix A) with near-twin pairs and creoles; source/transcription effects are not fully controlled. Direction and time are out of scope (the directed-layer paper). The cross-family comparison (§7) is deliberately structural; extending it to more families (e.g. Caucasian) and to a pre-registered areal test is future work.

10. Reproducibility

`make figure` redraws from bundled results (no corpus); `LEX_PATH=… make compute` regenerates the dissimilarity; `make analysis` reproduces §4–§6; `make compare SLUGS="ie an"` reproduces §7; `make` `controls` runs §8; `make manifest` writes Appendix A. Parameters (`MINSLOT`, `THR`, `MAXLANG`, `KNN`, feature subset) are environment-overridable in `src/compute_network.py`; permutation seeds are fixed. Versions: panphon, LingPy (LexStat), Lexibank, Glottolog 4.8 — pinned in `requirements.txt`. Code MIT; text, figures, data CC BY 4.0.

11. References

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Appendix A — Doculect manifest

The doculects, with Lexibank dataset, doculect ID, Glottocode, branch used for scoring, and concept count, are in `data/results/doculect_manifest_ie.csv` (and `_an.csv`; regenerate with `make manifest`). Provenance is exposed so source/transcription confounds can be audited (e.g. several near-twin systems share a source dataset).